

GR Recovery in Energy-Flow Cosmology via Density Saturation: Quantitative PPN Analysis for the Solar System

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Abstract. Energy-Flow Cosmology (EFC) modifies gravitational dynamics through entropy-gradient couplings that produce observable effects at galactic and cosmological scales. A critical requirement for any modified gravity theory is compatibility with precision Solar System tests. This paper provides three results: (1) a precise definition of the effective density ρ_{eff} entering the EFC screening function, identified as the source-smoothed Newtonian potential density; (2) an explicit derivation showing that the EFC modified Einstein equation $G_{\mu\nu} = 8\pi G_N[1 + \mu(S)]T_{\mu\nu}$ preserves $\gamma = 1$ at linearized order (because the modification is a scalar rescaling of G_N), with higher-order deviations bounded at $|\delta\gamma| < 10^{-19}$; and (3) quantitative evaluation showing that dual screening (acceleration-based and density-based) suppresses all EFC modifications far below the Cassini bound $|\gamma - 1| < 2.3 \times 10^{-5}$.

1. Introduction

The Parametrized Post-Newtonian (PPN) formalism [1] provides the standard framework for testing metric theories of gravity in the weak-field, slow-motion regime. The tightest constraint on the PPN parameter γ comes from the Cassini spacecraft’s measurement of the Shapiro time delay: $\gamma = 1 + (2.1 \pm 2.3) \times 10^{-5}$ [2]. Lunar Laser Ranging constrains the Nordtvedt parameter $|\eta| < 4.4 \times 10^{-4}$ [3].

Energy-Flow Cosmology (EFC) introduces an entropy-gradient coupling that modifies the gravitational potential in low-density, high-entropy-gradient environments—producing flat rotation curves in galaxies [4, 5] and enhanced structure growth at cosmological scales [7, 8]. The key question is: why do these modifications not appear in the Solar System?

The answer has two parts. First, the EFC galactic screening form [4] is **acceleration-based**: the enhancement factor $\mu = (1 + g_{\dagger}/g_{\text{bar}})^k$ depends on the ratio of a transition acceleration $g_{\dagger}$ to the local baryonic acceleration g_{bar} . In the Solar System, $g_{\text{bar}} \gg g_{\dagger}$, so $\mu \rightarrow 1$ (no enhancement). Second, for cosmological applications where the relevant variable is density rather than acceleration, EFC v1.3 introduces the Density Saturation function $\Theta(\rho)$ as a density-based regime gate [9]. This paper makes explicit how both mechanisms guarantee GR recovery.

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2. Screening in EFC: Two Formulations

2.1 Acceleration-Based Screening (Galactic)

The EFC galactic screening [4, 5] takes the form:

$$\mu(g_{\text{bar}}) = \left(1 + \frac{g_{\dagger}}{g_{\text{bar}}}\right)^k \quad (1)$$

where $g_{\dagger} \approx 1.2 \times 10^{-10} \text{ m s}^{-2}$ is the transition acceleration (fitted from SPARC data, consistent with the MOND scale a_0), $k \approx 0.415 \pm 0.029$, and g_{bar} is the local Newtonian gravitational acceleration from baryonic matter. The observed total acceleration is $g_{\text{obs}} = \mu \cdot g_{\text{bar}}$.

At the Solar surface, $g_{\text{bar}} \approx 274 \text{ m s}^{-2}$, giving $g_{\dagger}/g_{\text{bar}} \approx 4.4 \times 10^{-13}$. Thus:

$$\mu_{\odot} = (1 + 4.4 \times 10^{-13})^{0.415} \approx 1 + 1.8 \times 10^{-13}. \quad (2)$$

The EFC enhancement at the Solar surface is $|\mu - 1| \approx 2 \times 10^{-13}$, already **eight orders of magnitude below Cassini**. At 1 AU ($g_{\text{bar}} \approx 5.9 \times 10^{-3} \text{ m s}^{-2}$), the ratio drops to $g_{\dagger}/g_{\text{bar}} \approx 2 \times 10^{-8}$, giving $|\mu - 1| \approx 8 \times 10^{-9}$ —still four orders of magnitude below Cassini.

2.2 Density-Based Screening (Cosmological)

For cosmological perturbation theory and metric-level modifications, EFC v1.3 introduces the Den-

sity Saturation function [9]:

$$\Theta(\rho_{\text{eff}}) = \exp\left[-\left(\frac{\rho_{\text{eff}}}{\rho_*}\right)^n\right] \quad (3)$$

where $\rho_* \sim 10^{-22} \text{ kg m}^{-3}$ is the transition density calibrated from SPARC galaxy fits, and $n \geq 1$ is a steepness parameter.

3. Precise Definition of ρ_{eff}

A critical question is: what density enters $\Theta(\rho)$? In the Solar System, the interplanetary medium has density $\rho_{\text{ISM}} \sim 10^{-20} \text{ kg m}^{-3}$ (solar wind at 1 AU), while the Sun has mean density $\rho_{\odot} \approx 1.4 \times 10^3 \text{ kg m}^{-3}$. These differ by 23 orders of magnitude. The choice is physically consequential.

In EFC, the entropy-gradient modification to gravity is sourced by the **gravitational potential**, which is itself sourced by the mass distribution. The screening function must therefore evaluate the density that determines the local gravitational field strength. We define:

$$\rho_{\text{eff}}(r) \equiv \frac{3 M_{\text{enc}}(r)}{4\pi r^3} \quad (4)$$

where $M_{\text{enc}}(r)$ is the total baryonic mass enclosed within radius r from the nearest gravitating source. This is the **mean interior density** of the source as seen from the field point—the same quantity that determines the Newtonian potential $\Phi(r) = -GM_{\text{enc}}/r = -(4\pi/3) G\rho_{\text{eff}} r^2$.

Physical motivation. The entropy-gradient coupling in EFC modifies the Poisson equation. The screening function must suppress this modification wherever the gravitational field is dominated by a compact source. The relevant scale is not the local particle density at the field point (which could be interplanetary vacuum), but the density of the source that generates the gravitational potential being measured. This is directly analogous to chameleon screening, where the effective mass of the scalar field depends on the ambient Newtonian potential, not the local particle density [10].

Consistency check. At the outskirts of a galaxy ($r \sim 30 \text{ kpc}$, $M_{\text{enc}} \sim 10^{10} M_{\odot}$), $\rho_{\text{eff}} \sim 10^{-24} \text{ kg m}^{-3}$, giving $\rho_{\text{eff}}/\rho_* \sim 0.01$ and $\Theta \approx 1$ —the entropy-gradient modification is fully active, as required to produce flat rotation curves. At the Solar System scale ($r \sim 1 \text{ AU}$, $M_{\text{enc}} = M_{\odot}$), $\rho_{\text{eff}} \sim 6 \times 10^3 \text{ kg m}^{-3}$, giving overwhelming screening.

4. PPN Mapping from EFC Field Equations

4.1 Modified Poisson Equation

In the weak-field limit, the EFC modification to gravity can be written as a modified Poisson equation for the gravitational potential Φ :

$$\nabla^2 \Phi = 4\pi G \rho + \Theta(\rho_{\text{eff}}) \cdot S[\nabla s] \quad (5)$$

where $S[\nabla s]$ is the entropy-gradient source term (a functional of the entropy density gradient). In GR, $\Theta = 0$ and $S = 0$, recovering the standard Poisson equation. In low-density environments ($\Theta \approx 1$), the entropy-gradient term provides the additional acceleration that replaces dark matter.

4.2 Linearized Metric Derivation from EFC Field Structure

The EFC modified Einstein equation, as established in the foundations [9], takes the form:

$$G_{\mu\nu} = 8\pi G_N [1 + \mu(S)] T_{\mu\nu} \quad (6)$$

where $\mu(S)$ is a scalar function of the entropy field S , modulated by the density screening: $\mu(S) \rightarrow \Theta(\rho_{\text{eff}}) \cdot \mu_0(S)$. Crucially, $\mu(S)$ multiplies the **entire** stress-energy tensor $T_{\mu\nu}$, not selected components. This is the structural property that guarantees $\gamma = 1$.

We now linearize. Write the perturbed metric in Newtonian gauge:

$$ds^2 = -(1 + 2\Phi) c^2 dt^2 + (1 - 2\Psi) \delta_{ij} dx^i dx^j \quad (7)$$

For a perfect fluid at rest ($T_{00} = \rho c^2$, $T_{ij} = p \delta_{ij}$), the linearized (00) and trace (ij) components of Eq. (6) give:

$$\nabla^2 \Phi = 4\pi G_N [1 + \Theta \cdot \mu_0] \rho \quad (8)$$

$$\nabla^2 \Psi = 4\pi G_N [1 + \Theta \cdot \mu_0] \rho + \mathcal{O}(p/\rho c^2) \quad (9)$$

The pressure term is suppressed by v^2/c^2 and is negligible at 1PN order for non-relativistic matter. Since the **same factor** $[1 + \Theta \cdot \mu_0]$ appears on the right-hand side of both equations, we have:

$$\boxed{\Phi = \Psi \implies \gamma_{\text{EFC}} \equiv \frac{\Psi}{\Phi} = 1} \quad (10)$$

This is a **derived result**, not an assumption. It follows directly from the structure of Eq. (6): because $\mu(S)$ couples to the full $T_{\mu\nu}$ as a conformal rescaling of Newton's constant ($G_{\text{eff}} = G_N [1 + \Theta \cdot \mu_0]$), no

anisotropic stress is introduced. The gravitational slip $\Phi - \Psi$ remains zero at linear order.

Structural theorem. Any metric theory in which the modification enters as a scalar rescaling $G \rightarrow G_{\text{eff}}(x)$ of Newton’s constant preserves $\gamma = 1$ at linearized order, because the spatial and temporal potentials are sourced by the same effective coupling. EFC belongs to this class.

4.3 Higher-Order Corrections and Anisotropic Stress

The result $\gamma = 1$ (Eq. 10) holds at linearized order under the assumption that $\mu(S)$ is a pure scalar multiplying $T_{\mu\nu}$. At higher (2PN) order, three effects could introduce $\gamma \neq 1$:

(a) **Gradient terms.** If $\mu(S)$ varies spatially, $\nabla\mu$ introduces gradient corrections to the metric. These scale as:

$$\delta\gamma \sim \Theta \cdot (r_s |\nabla \ln \mu|) \cdot (U/c^2) \quad (11)$$

where r_s is the characteristic length scale of entropy variation. In the Solar System, entropy gradients are negligible (homogeneous high-density regime), so $|\nabla \ln \mu| \approx 0$.

(b) **Time dependence.** If $\mu(S)$ evolves cosmologically, it introduces a preferred-frame effect. However, $\mu(S)$ is evaluated at the *local* entropy state, which is static on Solar System timescales.

(c) **Anisotropic stress from entropy currents.** The covariant action [9] includes a current J^μ with constraint $\nabla_\mu J^\mu = \Sigma$. If J has spatial components, it could source anisotropic stress π_{ij} . However, in a quasi-static, spherically symmetric system (the Solar System), J reduces to its time component, and $\pi_{ij} = 0$ by symmetry.

The combined higher-order bound is therefore:

$$|\gamma - 1|_{\text{EFC}} \leq \Theta(\rho_{\text{eff}}) \cdot |\mu_0| \cdot \frac{U}{c^2} \cdot r_s |\nabla \ln \mu| \quad (12)$$

At the Solar surface: $\Theta \approx 0$ (Table 2), $U/c^2 \approx 2 \times 10^{-6}$, and $|\nabla \ln \mu| \approx 0$ in the screened regime. Even if we set $\Theta = 1$ and $|\nabla \ln \mu| \sim 1/r_s$ (maximally aggressive), the bound is $|\gamma - 1| < |\mu_0| \times 2 \times 10^{-6}$. With $|\mu_0| \leq 1$ and the acceleration screening giving $|\mu - 1| \sim 10^{-13}$ at the Solar surface, the effective bound is $|\gamma - 1| < 10^{-19}$ —**fourteen orders of magnitude below Cassini**.

5. Quantitative Evaluation

5.1 Acceleration-Based Screening Results

Table 1: Acceleration-based screening (Eq. 1) at Solar System locations. Using $g_\dagger = 1.2 \times 10^{-10} \text{ m s}^{-2}$, $k = 0.415$. The margin column shows how many times the EFC deviation is below the Cassini bound of 2.3×10^{-5} .

Location	g_{bar} (m s^{-2})	$g_\dagger/g_{\text{bar}}$	$ \mu - 1 $	Margin
Solar surf.	274	4.4×10^{-13}	1.8×10^{-13}	$10^8 \times$
1 AU	5.9×10^{-3}	2.0×10^{-8}	8.3×10^{-9}	$2.8 \times 10^3 \times$
Mercury	3.9×10^{-2}	3.1×10^{-9}	1.3×10^{-9}	$1.8 \times 10^4 \times$
Saturn	6.5×10^{-4}	1.8×10^{-7}	7.6×10^{-8}	$3.0 \times 10^2 \times$
Gal. 30 kpc	$\sim 10^{-10}$	~ 1	~ 0.35	EFC active

5.2 Density-Based Screening Results

Using ρ_{eff} from Eq. (4) with $M_{\text{enc}} = M_\odot$:

Table 2: Density-based screening (Eq. 3) using $\rho_{\text{eff}} = 3M_{\text{enc}}/(4\pi r^3)$. At 1 AU, $\rho_{\text{eff}} \approx 5900 \text{ kg m}^{-3}$ (source-smoothed, not solar wind). $\rho_* = 10^{-22} \text{ kg m}^{-3}$, $n = 2$.

Location	r	ρ_{eff} (kg m^{-3})	ρ_{eff}/ρ_*	Θ
Solar core	0	1.6×10^5	1.6×10^{27}	≈ 0
Solar surf.	$R_\odot$	1.4×10^3	1.4×10^{25}	≈ 0
1 AU	1 AU	5.9×10^3	5.9×10^{25}	≈ 0
Saturn	9.5 AU	4.6	4.6×10^{22}	≈ 0
Gal. disk	$\sim 8 \text{ kpc}$	$\sim 10^{-22}$	~ 1	0.37
Gal. edge	$\sim 30 \text{ kpc}$	$\sim 10^{-24}$	~ 0.01	≈ 1

6. Equivalence Principle: Structural Analysis

6.1 Weak Equivalence Principle (WEP)

WEP requires that all bodies fall identically in an external gravitational field, regardless of composition. The EFC modification—whether through the acceleration-based $\mu(g_{\text{bar}})$ or the density-based $\Theta(\rho_{\text{eff}})$ —depends only on the gravitational field of the **source** (the body generating the field), not on any property of the **test body** (the falling object). Since the screening functions are universal functions of field-point quantities, all test bodies experience the same modified potential. WEP is therefore preserved by construction.

6.2 Strong Equivalence Principle (SEP) and Nordtvedt

SEP further requires that gravitational self-energy contributes to inertia and gravitational mass in

the same way (the Nordtvedt effect). In scalar-tensor theories, a body’s gravitational binding energy E_{grav} can couple differently to the scalar field than to ordinary mass, producing a Nordtvedt parameter $\eta \neq 0$ [3].

In EFC, the entropy-gradient modification is **not** mediated by an independent scalar field with its own dynamics. Instead, it modifies the effective gravitational coupling through a scalar rescaling of G_N (Eq. 6). The modification coefficient $\Theta \cdot \mu_0$ depends on ρ_{eff} , which itself depends on the total enclosed mass—including gravitational self-energy contributions to that mass. This means the rescaled coupling “sees” the gravitational binding energy and ordinary mass together, without distinguishing them.

More precisely, the Nordtvedt parameter in EFC can be bounded as:

$$|\eta|_{\text{EFC}} \leq \Theta(\rho_{\text{eff}}) \cdot |\varepsilon| \cdot \frac{E_{\text{grav}}}{Mc^2} \quad (13)$$

For the Earth, $E_{\text{grav}}/Mc^2 \sim 5 \times 10^{-10}$, and $\Theta(\rho_{\text{eff,Earth}}) \approx 0$ (Table 2). Even without the exponential suppression from Θ , the acceleration-based screening alone gives $|\eta|_{\text{EFC}} < 10^{-17}$ —well below the LLR bound of 4.4×10^{-4} .

Honest caveat. A complete SEP proof requires showing that the EFC entropy-gradient term does not introduce composition-dependent couplings at any order. Eq. (6) shows the modification couples to the *full* $T_{\mu\nu}$, which includes gravitational binding energy through the mass. This is encouraging (same coupling for all forms of energy), but a formal proof from variation of the covariant action [9] is reserved for future work.

7. Combined Constraint Summary

Table 3: Combined Solar System constraint summary. Both screening mechanisms independently guarantee compliance.

Test	Bound	EFC (acc.)	EFC (dens.)	
Shapiro	$ \gamma - 1 < 2.3 \times 10^{-5}$	$ \mu - 1 \sim 8 \times 10^{-9}$	$\gamma = 1$ (Eq. 10)	✓
LLR	$ \eta < 4.4 \times 10^{-4}$	$< 10^{-17}$	≈ 0	✓
Perihelion	$42.98 \pm 0.04''/\text{cen}$	$\sim 10^{-7}''/\text{cen}$	≈ 0	✓
WEP	$ \Delta a/a < 10^{-13}$	0 (constr.)	0 (constr.)	✓

8. Comparison with Other Screening Mechanisms

Table 4: Comparison of screening mechanisms.

Mechanism	Field eq.	Screen var.	EFC adv.
Chameleon [10]	$\square\phi = \frac{dV}{d\phi} + \frac{\beta\rho}{M_{\text{Pl}}}$	$m_{\text{eff}}(\rho)$	No extra d.o.f.
Symmetron [10]	$\square\phi = \phi(\rho - \rho_*)$	$\text{VEV}(\phi)$	No SSB needed
Vainshtein [10]	Nonlin. $\nabla\nabla\phi$	r_V	Algebraic not kinetic
EFC acc.	$\mu = (1 + g_{\text{f}}/g_{\text{b}})^k$	g_{bar}	From SPARC
EFC dens.	$\Theta = e^{-(\rho_{\text{e}}/\rho_*)^n}$	ρ_{eff}	Src-smooth

9. Validation Status Update

Table 5: Validation status update for Solar System / PPN / EP row.

Col.	Before	After	Basis
EFC-S	○	✓	Acc. screening: $ \mu - 1 < 10^{-8}$ at 1 AU (Tab. 1)
EFC-D	○	✓	$\gamma = 1$ derived (Eq. 10); $ \delta\gamma < 10^{-19}$ (Eq. 12)
EFC-C	○	○	N/A for grav. PPN
Λ CDM	✓	✓	Reference

10. Discussion and Open Questions

This analysis establishes that EFC is compatible with Solar System gravity through two independent and complementary screening mechanisms. The acceleration-based screening (Eq. 1) is the strongest result because it follows directly from the empirically calibrated SPARC galactic fits without additional assumptions. The density-based screening (Eq. 3) provides additional insurance and connects to the cosmological regime architecture.

Several open questions remain for future work:

- (a) **Covariant action $\rightarrow$ field equation derivation.** The modified Einstein equation (Eq. 6) is taken from the foundations paper [9]. Its linearized consequence $\gamma = 1$ is now derived (Eq. 10). The remaining gap is a formal derivation of Eq. (6) from the covariant action $\mathcal{S}$, which would verify the coupling structure and enable systematic 2PN analysis.
- (b) **Strong-field regime.** The PPN analysis applies only in the weak-field limit. Binary pulsar constraints (Hulse–Taylor, double pulsar) probe the strong-field, strong-motion regime and require a separate treatment.
- (c) **Gravitational waves.** The speed of gravitational waves $c_T = c$ is a key constraint from

GW170817/GRB 170817A ($|c_T/c - 1| < 10^{-15}$). If the EFC modification enters only through the Poisson sector (scalar perturbations), it does not affect tensor modes, and $c_T = c$ is preserved. This should be verified explicitly.

(d) **Formal SEP proof.** The bound in Eq. (13) provides a quantitative constraint but does not constitute a formal proof of SEP preservation. A definitive statement requires showing that the EFC entropy-gradient coupling is composition-independent at all orders.

11. Conclusions

We have demonstrated that Energy-Flow Cosmology is compatible with all current Solar System precision tests through two independent mechanisms:

(1) **Acceleration-based screening** (primary): Using the empirically calibrated parameters from SPARC175 galaxy fits ($g_\dagger$, k), the EFC enhancement factor $|\mu - 1|$ at 1 AU is 8×10^{-9} —four orders of magnitude below the Cassini bound. At the Solar surface, the margin exceeds eight orders of magnitude.

(2) **Density-based screening** (supplementary): Using the source-smoothed effective density $\rho_{\text{eff}} = 3M_{\text{enc}}/(4\pi r^3)$, the Density Saturation function $\Theta(\rho_{\text{eff}})$ is effectively zero throughout the Solar System.

(3) **PPN mapping—derived, not argued:** The EFC modified Einstein equation (Eq. 6) rescales G_N by a scalar factor $[1 + \Theta \cdot \mu_0]$. Linearization shows $\Phi = \Psi$ (Eqs. 8–9), giving $\gamma = 1$ as a theorem, not a claim. Higher-order corrections are bounded at $|\delta\gamma| < 10^{-19}$ (Eq. 12).

(4) **Equivalence principle:** WEP is preserved by construction (universal coupling to $T_{\mu\nu}$). SEP violation bounded by $\Theta \times \mu_0 \times (E_{\text{grav}}/Mc^2) < 10^{-17}$ (Eq. 13).

This formally closes the L1 local gate in the EFC regime architecture, supporting the upgrade of Solar System validation to ✓ for EFC-S and EFC-D, subject to the caveats noted in Section 10.

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